

Academic appointments

September 2024 — **Toyota Technological Institute at Chicago**
Present Research Assistant Professor

Education

September 2018 — **University of Toronto**, Toronto, ON
June 2024 Ph.D. in Computer Science
Thesis title: *Natural Language Processing for Slang*
Thesis advisor: *Yang Xu*

August 2016 — **Georgia Institute of Technology**, Atlanta, GA
May 2018 M.S. in Computer Science
Specialization in Machine Learning

September 2012 — **University of Waterloo**, Waterloo, ON
June 2016 B.S. in Computer Science
Minor in Economics

Honors and awards

- Queen Elizabeth II Graduate Scholarship in Science & Technology (2022)
- Cecil and Edna Cotton Scholarship (2015)
- University of Waterloo - President's Research Award (2015)
- University of Waterloo - President's Scholarship of Distinction (2012)
- Bronze Award - Canadian Computing Competition Stage Two (2011)

Publications

10. **Domain-Specific Jargon in Large Language Models: A Comparative Analysis between General-Purpose and Specialist Models**

Darin Keng and *Zhewei Sun*

To appear in 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026), Long Paper.

9. **Measuring, Localizing, and Reducing the AI-Like Writing Style of Post-Trained LLMs**

Aniket Anand, Janvijay Singh, *Zhewei Sun*, Dilek Hakkani-Tür, and Nick Feamster

To appear in 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026), Long Paper.

8. **From Utterances to Networks: Modelling Slang Adoption and Diffusion Across Subreddits**

Xiaoning Wang, Ted Underwood, and *Zhewei Sun*

To appear in 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026), Long Paper.

7. **How do Language Models Generate Slang: A Systematic Comparison between Human and Machine-Generated Slang Usages**

Siyang Wu and *Zhewei Sun*

In Findings of the Association for Computational Linguistics: EMNLP 2025, Long Paper, Poster.

6. **Toward Informal Language Processing: Knowledge of Slang in Large Language Models**

Zhewei Sun, Qian Hu, Rahul Gupta, Richard Zemel, and Yang Xu

In Proceedings of the 2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2024), Long Paper, Oral.

5. **Tracing Semantic Variation in Slang**

Zhewei Sun and Yang Xu

In Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP 2022), Long Paper, Oral.

4. **Semantically Informed Slang Interpretation**

Zhewei Sun, Richard Zemel, and Yang Xu

In Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics - Human Language Technologies (NAACL-HLT 2022), Long Paper, Oral.

3. **A Computational Framework for Slang Generation**

Zhewei Sun, Richard Zemel, and Yang Xu

Transactions of the Association for Computational Linguistics, 9, 462-478 (TACL 2021), Oral Presentation at ACL 2021.

2. Slang Detection and Identification

Zhengqi Pei, *Zhewei Sun*, and Yang Xu

In Proceedings of the 2019 Conference on Computational Natural Language Learning (CoNLL 2019), Long Paper, Poster.

1. Slang Generation as Categorization

Zhewei Sun, Richard Zemel, and Yang Xu

In Proceedings of the 41st Annual Meeting of the Cognitive Science Society (CogSci 2019), Poster.

Working papers

1. On Cross-Lingual Regularities in Slang

Kefan Yu, Nolan Pozzobon, and *Zhewei Sun*

Under review.

Teaching

April 2026 —
July 2026

Toyota Technological Institute (TTI-Japan)

Co-Instructor

Machine Learning II - Spring 2026

- Co-instructor for 4th year undergraduate course focusing on deep learning. Lectures, homework, and exams were delivered in English.

September 2018 —
December 2023

University of Toronto - Department of Computer Science

Teaching Assistant

Courses:

- CSC485/2501 - Computational Linguistics (Fall 2018, Fall 2019, Fall 2020, Fall 2021, Fall 2022, Fall 2023).
- CSC401/2511 - Natural Language Computation (Winter 2019, Winter 2020, Winter 2021).
- CSC384 - Introduction to Artificial Intelligence (Summer 2019, Summer 2020, Summer 2021, Summer 2022, Summer 2023).

August 2017 —
May 2018

Georgia Institute of Technology - School of Interactive Computing

Graduate Teaching Assistant

Courses:

- CS7646 - Machine Learning for Trading (Fall 2017).
- CS7650 - Natural Language Processing (Spring 2018).

Experience

- September 2018 — June 2024* — **University of Toronto - Computational Linguistics Group**
Research Assistant
- Advisor: *Yang Xu*
Project: Natural language processing for informal language, with specific focus on slang.
- August 2017 — May 2018* — **Georgia Institute of Technology - Computational Linguistics Lab**
Graduate Research Assistant
- Advisor: *Jacob Eisenstein*
Project: Discourse neural machine translation and language modelling.
- August 2016 — December 2016* — **Georgia Institute of Technology - Sonification Lab**
Research and Development Studio
- Advisor: *Bruce Walker*
Project: Heads-up displays in driving simulators.
- September 2015 — December 2015* — **University of Waterloo - Department of Computer Science**
CS Undergraduate Research Assistant
- Advisor: *Justin W. L. Wan*
Project: Computational finance on GPU systems.
- May 2013 — August 2013* — **IBM Canada**
Software Engineering Intern
- Supervisor: *Christopher Kaufman*
Project: Score-carding feature development for the Cognos TM1 business analytics suite.

Invited talks & posters

- June 2026* — **Toyota Technological Institute (TTI-Japan)**
Title: How do language models generate slang: A systematic comparison between human and machine-generated slang usages

<i>April 2026</i>	University of Illinois at Urbana-Champaign - Midwest Speech and Language Days (MSLD) 2026 Title: From Utterances to Networks: Modelling Slang Adoption and Diffusion Across Subreddits
<i>April 2026</i>	University of Illinois at Urbana-Champaign - Midwest Speech and Language Days (MSLD) 2026 Title: How do language models perceive creative language: A deep dive into generation and multilingual adaptation of novel slang
<i>April 2025</i>	University of Notre Dame - Midwest Speech and Language Days (MSLD) 2025 Title: Natural Language Processing for Slang
<i>February 2025</i>	University of Chicago - CMSC 25700 Guest Lecture Title: Data vs. Knowledge: Two Approaches to Enable NLP for Slang
<i>March 2024</i>	State University of New York at New Paltz Title: Contextualizing Natural Language Agents: The Case of Slang
<i>February 2024</i>	University of Lethbridge Title: Contextualizing Natural Language Agents: The Case of Slang
<i>February 2021</i>	Vector Institute Research Symposium 2021 Title: A Computational Framework for Slang Generation
<i>September 2020</i>	Vector Institute Natural Language Processing Symposium Title: Informal Natural Language Processing: The Case of Slang
<i>July 2020</i>	The 42nd Annual Meeting of the Cognitive Science Society Title: The Emergence and Propagation of Online Slang
<i>October 2019</i>	Vector Institute Natural Language Processing Symposium Title: Informal Natural Language Processing: The Case of Slang
<i>September 2019</i>	Google Brain Toronto Title: Informal Natural Language Processing: The Case of Slang

Internal service

- Co-organizer, UChicago/TTIC Communications & Intelligence seminar, Spring 2026
- Faculty Judge, TTIC Student Workshop 2026

- Faculty Supervisor, UChicago/TTIC joint NLP seminar, Winter 2025
- Co-organizer, UChicago/TTIC joint NLP seminar, Fall 2024
- Webmaster, TTIC Speech and Language group, 2024 - Present.

External service

- Reviewer, ACL Rolling Review (October 2023 - Present), 47 submissions
- Reviewer, COLM 2025, 2026, 5 submissions
- Reviewer, CogSci 2020, 2022, 4 submissions
- Reviewer, TACL, 1 submission
- Reviewer, PLOS One, 4 submissions
- Reviewer, Computers in Human Behavior: Artificial Humans, 1 submission
- Reviewer, Journal of Applied Science and Engineering, 1 submission

Students advised

- Darin Keng (June 2025 - Present)
Primary master's thesis co-advisor.
Master's student, University of Chicago.

Students mentored

- Vivienne Zhang (June 2026 - Present)
Undergraduate student, University of Chicago.
- Aniket Anand (May 2026 - Present)
Ph.D. student, University of Chicago.
- Kefan Yu (June 2025 - Present)
Undergraduate student, Northwestern University.
Now M.S. Student at University of Tokyo.
- Xiaoning Wang (January 2025 - Present)
Ph.D. student, University of Illinois Urbana-Champaign.
- Marcelo Beramendi Caballero (October 2024 - Present)
Ph.D. student, Toyota Technological Institute at Chicago.
- Siyang Wu (January 2025 - October 2025)
Master's student, University of Chicago.
Now Ph.D. Student at University of Chicago.
- Nolan Pozzobon (January 2025 - May 2025)
Undergraduate student, University of Chicago.
Now M.S. Student at Carnegie Mellon University.
- Zhenyu Wang (May 2022 - August 2022)
Undergraduate student, University of Toronto.
Now M.S. Student at University of Oxford.

- Sherry Yuan (May 2020 - August 2020)
Undergraduate student, University of Toronto.
- Yishan Jiang (September 2019 - April 2020)
Master's student, University of Toronto.
Now at Google.
- Zhengqi Pei (September 2018 - August 2019)
Undergraduate student, University of Toronto.
Now M.S. Student at University of the Chinese Academy of Sciences.

Open-source software

- SlangGen. Natural language processing pipeline to generate novel slang usages based on historical slang usages from dictionaries.
<https://github.com/zhewei-sun/slanggen>
- CatGO. A light-weight Python implementation of popular categorization models with automatic parameter optimization.
<https://github.com/zhewei-sun/CatGO>